Does Functional Similarity Induced by Distillation Preserve Internal Representations and Explanatory Mechanisms?

Martín L. Perciante
Dpto. de Informática e IA
Universidad Católica del Uruguay
Montevideo, Uruguay
martinluca.perciante@ucu.edu.uy

Fiorella Cravero
Dpto. de Informática e IA
Universidad Católica del Uruguay
Montevideo, Uruguay
fiorella.cravero@ucu.edu.uy

J. Matías Di Martino
Dpto. de Informática e IA
Universidad Católica del Uruguay
Montevideo, Uruguay
matias.dimartino@ucu.edu.uy

***Abstract*—Knowledge distillation is often evaluated in terms of functional similarity between teacher and student models, measured through predictive performance. However, it remains unclear whether this functional similarity also implies the preservation of internal representations and explanatory mechanisms without explicitly imposing them. This work investigates this question in autoregressive Transformers. A TinyLlama teacher is distilled into a family of half-depth (11-layer) students, each under a different training configuration. Internal representations are compared through layer-wise attention-map metrics, while explanatory mechanisms are evaluated through LIME, SHAP, and Integrated Gradients, moving beyond qualitative examples to aggregate agreement and deletion-based faithfulness metrics. The study is conducted on binary IMDb sentiment classification and on four-class AG News topic classification. Results show that distillation does not transfer the teacher's explanatory mechanisms on either dataset. Representational similarity, in turn, is only partially and inconsistently attributable to distillation: attention-level metrics show that a non-distilled baseline is as close to the teacher as the distilled students. These findings suggest that the observed similarity to the teacher is driven more by the shared data and task than by distillation itself, indicating that functional similarity does not, by itself, guarantee the preservation of a teacher's explanatory mechanisms or internal representations.**

***Index Terms*—Distillation, Attention, XAI, Explainability**

I. Introduction

Explainable Artificial Intelligence (XAI) seeks to interpret and explain the behavior of complex models, essential in domains where deployed systems must justify autonomous decisions and foster user trust. This task is particularly difficult for Large Language Models (LLMs), where probabilistic text generation replaces classification with a single class label output. Transformer-based models are the architecture of choice for this kind of modeling in current Natural Language Processing (NLP), driving recent progress largely by growing in parameters and complexity; the resulting computational cost has motivated compression techniques such as knowledge distillation (KD), in which a smaller model (student) is trained to approximate the output distribution of a larger model (teacher) [1], alongside dataset training.

This research was funded by the Agencia Nacional de Investigación e Innovación (ANII), Uruguay, grant number POS_NAC_2025_4_189264.

Logit-based KD is centered on functional similarity: the same input is made to produce a similar output, regardless of the internal representation that produces it. Prior work has shown that different attention configurations can induce similar functional behavior [2], raising the question of whether KD enforces a preference for the teacher's internal representations, or whether it can converge to different attention maps while reproducing the teacher's output distribution. Bringing XAI into this picture, different models can provide different explanations for the same example while achieving similar performance over a dataset.

This work investigates whether the functional similarity induced by logit-based KD extends to similarity in internal representations and explanatory mechanisms. We adopt a generative classification setting, in which target classes correspond to single vocabulary tokens, allowing internal representations to be compared directly between teacher and student. We conducted the experiments on binary **IMDb** sentiment classification and on four-class **AG News** topic classification.

This work's contributions are:

- We show that logit-based KD, even when it reproduces the teacher's output distribution with increasing fidelity, does not correspondingly transfer its explanatory mechanisms.
- We show that representational similarity to the teacher (attention geometry) is not a reliable signature of distillation itself.
- We show this dissociation holds across both a binary (**IMDb**) and a four-class (**AG News**) classification task, indicating it is not an artifact of a single dataset or label space.

II. Related Work

KD transfers information from a high-capacity teacher to a smaller student by adding a *Kullback–Leibler divergence* (D_{KL}) term to the loss function, often softened by a temperature T [1]. KD is widely used in deep networks and large language models, but its success is evaluated almost exclusively via functional metrics such as accuracy [3]. Some works explicitly bias the loss towards representational alignment, such as Attention Distillation [4], which directly supervises student attention maps against the teacher's. Unlike Attention Distillation, we test whether such alignment emerges implicitly from logit-based KD, without any explicit representational objective.

Post-hoc explainability methods such as **LIME** [5] and **SHAP** [6] treat the model as a black box, while gradient-based methods such as **Integrated Gradients** [7] exploit differentiability; all three are commonly used to quantify token-level contributions to a prediction. Beyond post-hoc attribution, attention maps offer a potentially interpretable internal mechanism, but whether they constitute a faithful explanation remains contested: Jain and Wallace [2] show that alternative attention configurations can induce similar predictions, whereas Wiegreffe and Pinter [8] argue attention can still provide plausible explanations under appropriate conditions. Given this contested status, we treat attention maps and post-hoc attributions as complementary probes of whether KD implicitly transfers a teacher's explanatory mechanisms.

III. Experiments

A. Experimental Setup

The project used **Python**, **PyTorch**, Hugging Face's **transformers**, **lime/shap** for explanation, **SciPy** for the rank/value-correlation statistics used in the aggregate explanation metrics, and **scikit-learn** for evaluation metrics.

The teacher is **TinyLlama-1.1B-Chat-v1.0** [9], a 22-layer decoder-only Transformer [10]. Every student shares the teacher's embeddings, attention mechanism, and hidden dimensions, reducing only depth to 11 randomly initialized blocks. This 2:1 Transformer layer correspondence allows for a clean comparison between teacher and student models, yielding a compression rate $CR \approx 1.787$ (approximately a 44% size reduction).

The studied problem is reformulated as a generative classification [11], where class-labels are single vocabulary tokens. Dataset inputs are inserted into a fixed prompt template, and only the class-token logits are compared at inference. For **IMDb** [12] (50000 reviews, labels *positive/negative*), the template is `"Review: {input} Summary:"`; filtered to the teacher's 2048-token context, splits are 22492/2500/24995 (train/val/test). For **AG News** [13] (4 classes: *World*, *Sports*, *Business*, *Sci/Tech*), the template is `"Article: {input} Theme:"`. Since *Sci/Tech* tokenizes to four sub-tokens, it is relabeled *Science* (one token) purely to preserve the single-token framing. Splits are 30000/100/7600 (balanced). All other hyperparameters are shared across datasets.

The teacher is fully fine-tuned per dataset for one epoch (learning rate 2×10^{-5}, AdamW, early stopping after 5 non-improving validation rounds) over the cross-entropy loss ($\mathcal{L}_{CE}$) [14]. Every student is randomly initialized and trained for 3 epochs (learning rate 1×10^{-4}, no early stopping), minimizing (1), where T is the training temperature, p_T and p_S are the teacher and student probability distributions over the complete token vocabulary, and $p_T^{(c)}$ and $p_S^{(c)}$ are the teacher and student probability distributions restricted to the class-label tokens. The hyperparameter α balances $\mathcal{L}_{CE}$ against the full-vocabulary distillation term, while β weights an additional distillation term restricted to the class-label tokens. A temperature $T = 2$ was fixed for every training process.

$$\begin{aligned}\mathcal{L} = \alpha\,\mathcal{L}_{CE} + (1-\alpha)\,T^2 D_{KL}\left(p_T \| p_S\right) \\ + \beta\,T^2 D_{KL}\left(p_T^{(c)} \| p_S^{(c)}\right)\end{aligned} \tag{1}$$

Both teacher P and baseline B were trained with $\alpha = 1$, $\beta = 0$. B therefore receives no signal from the teacher and shares the reduced architecture with the distilled students, serving as a non-distilled reference. S_0 ($\alpha = 0.9$, $\beta = 0$) is dominated by $\mathcal{L}_{CE}$, whereas S_1 ($\alpha = 0.5$, $\beta = 0$) balances both terms from the dataset and the teacher. This configuration of settings allows us to compare KD's impact on the results. Additionally, a third student S_2 ($\alpha = 0.5$, $\beta = 0.25$) keeps the same balance between terms, but adds the class-restricted term, reinforcing agreement with the teacher over the class-label tokens specifically and helping preserve its argmax there.

Explanations from **LIME**, **SHAP**, and **Integrated Gradients** (zero-embedding baseline) are compared between teacher and student via aggregate metrics, computed over $N = 100$ held-out examples per dataset for all three methods. Representational comparisons use the 2:1 teacher-student layer correspondence: attention maps are averaged over held-out samples ($N = 200$) at each layer, and student layer i is then compared against $P_{\text{avg},i} = (P_{2i} + P_{2i+1})/2$, the average of teacher layers $2i$ and $2i{+}1$. All experiments ran on an NVIDIA RTX 5070 Ti (16GB).

B. Comparison Metrics

We compare attention maps with complementary geometric, probabilistic, and spectral measures. Given two attention matrices A, B and their vectorizations $\vec{A}, \vec{B}$, we compared them via *cosine similarity* $\cos(\theta) = \langle \vec{A}, \vec{B} \rangle / (\|\vec{A}\|_2 \|\vec{B}\|_2) \in [-1, 1]$, the *Frobenius distance* $\|A - B\|_F = \sqrt{\sum_{i,j} (A - B)_{ij}^2}$ and the row-averaged *Jensen–Shannon divergence* $D_{JS}(A\|B) = \frac{1}{n} \sum_i D_{JS}(A_i\|B_i)$ where A_i, B_i are the i-th rows of A, B and $D_{JS}(A_i\|B_i) = \frac{1}{2} D_{KL}(A_i\|M_i) + \frac{1}{2} D_{KL}(B_i\|M_i)$ with $M_i = (A_i + B_i)/2$. For $n \times n$ attention maps, the *Frobenius distance* is bounded by $\sqrt{2n}$ (64 for our $n = 2048$ maps). D_{JS} is theoretically bounded in $[0, \ln 2]$ if measured in nats [15], where 0 means equal distributions. For spectral metrics we propose the entropy-based *effective rank* (ER) [16] and the *participation ratio* (PR) [17].

For evaluating explainability methods, we compute aggregate metrics over held-out samples, comparing each student's word-level attribution against the teacher's for every method. Metrics include *Jaccard@k* $= |A_k \cap B_k| / |A_k \cup B_k|$, where A_k, B_k are the top-k attributed word sets of two models ($k = 10$), *Spearman correlation* of signed scores over the common attributed vocabulary, and deletion-based *faithfulness@k* [18], the drop in true-class probability after masking the top-k words

Table I: Accuracy, F1 and average D_{KL} to the teacher P at $T = 1$ and $T = 2$ (**IMDb**: $N = 3000$ balanced subset; **AG News**: full test set). The **best** and <u>second best</u> results are highlighted.

Model	Acc.	F1	D_{KL} $T=1$	D_{KL} $T=2$
IMDb				
P	**0.954**	**0.954**	–	–
B	0.857	0.857	2.847	1.686
S_0	0.813	0.810	**0.103**	<u>0.131</u>
S_1	0.798	0.797	<u>0.104</u>	**0.127**
S_2	<u>0.867</u>	<u>0.867</u>	0.111	0.192
AG News				
P	0.848	0.847	–	–
B	**0.886**	**0.886**	2.080	1.784
S_0	<u>0.883</u>	<u>0.883</u>	0.629	0.418
S_1	0.878	0.878	<u>0.278</u>	**0.228**
S_2	0.880	0.880	**0.247**	<u>0.246</u>

(computed per model, including the teacher, as reference). Given **LIME's** stochastic nature, we also measure *stability* as the Jaccard@10 mean across 5 re-runs with different seeds.

C. Evaluation on the Test Set

Table I reports classification performance and average D_{KL} to the teacher, at $T = 2$ (training temperature) and at $T = 1$ (inference temperature), for all models on both datasets. On **IMDb**, distillation's effect on accuracy is mixed: only the class-restricted S_2 edges out the non-distilled B (0.867 vs. 0.857), while S_0 and S_1 both *underperform* B (0.813 and 0.798). On **AG News**, this pattern is more pronounced: B achieves the *highest* accuracy of all five models ($B > S_0 > S_2 > S_1 > P$). D_{KL}, by contrast, is markedly higher for B than for any distilled student on both datasets and temperatures. The ordering among S_0, S_1, and S_2 is less stable: it roughly tracks distillation strength on **AG News** ($S_0 > S_1 \gtrsim S_2$), but reverses on **IMDb**, where S_2 is consistently the most divergent of the three despite its additional class-restricted term, attaining *higher* D_{KL} than S_1 despite better accuracy: global D_{KL} is not a predictive proxy for classification performance, nor, as shown below, of explanatory or representational alignment.

D. Functional Evaluation and Explanation Comparison

LIME explanations for a randomly chosen **AG News** example show the teacher's top attributions concentrated on genuine topic-relevant words; students share most of these words, mixing in generic phrase tokens absent from the teacher's list. A similar pattern holds on **IMDb**, where the teacher and most students concentrate on genuine sentiment words, except S_1, which consistently favors function words over content words regardless of its classification performance.

SHAP explanations for the same examples are far less concentrated: the top-10 words capture only a small fraction of the total attribution, with the remaining ~140 tokens accounting for most of it, and content/function words appear interspersed across all five models rather than following a clean teacher/student split. **Integrated Gradients**, for every model including the teacher, is instead dominated by one or two extreme-magnitude spikes on semantically arbitrary tokens (proper nouns, subword fragments) rather than topical or sentiment content.

Table II reports aggregate **LIME** agreement (vs. teacher), faithfulness, and stability (mean pairwise Jaccard@10 across 5 re-runs, $N = 20$) on both datasets, $N = 100$, $k = 10$. On agreement and faithfulness, S_2 is consistently the best-aligned and most-faithful student, though the gap to B is modest on agreement but larger on faithfulness. Stability follows a less clean pattern: the teacher is more stable than the students on **AG News**, but the least stable of all on **IMDb**, so stability alone is not a proxy for explanation quality.

SHAP (Table II) shows the same S_2-best ranking, but faithfulness near zero for *every* model, including the teacher. Since this holds uniformly across all models, it cannot be a distillation effect; whether it stems from the deletion-based faithfulness metric itself or from the specific words **SHAP** attributes importance to remains an open question.

Integrated Gradients (Table II) shows high top-10 token overlap ($\approx$ 0.78–0.89) alongside near-zero or negative rank correlations, indicating agreement on *which* tokens matter without agreement on their relative importance.

Across both datasets and every model, including the teacher, **LIME** attains the highest faithfulness of the three methods (Table II). This ranking may partly reflect an alignment between method and metric rather than a genuine difference in explanation quality: **LIME** builds its attributions by masking the input itself, the same operation deletion-based faithfulness evaluates, whereas **SHAP** attributes importance relative to a reference distribution and **Integrated Gradients** relative to input gradients along a path, neither of which directly perturbs the input being scored.

E. Representational Comparison

Fig. 1 reports *cosine similarity*, *Frobenius distance*, and D_{JS} between each student's attention maps and P_{avg} across all layers per dataset. It also shows PR and ER values for every layer of every model. On **IMDb**, B has a slightly lower *cosine similarity* to P_{avg} than the three distilled variants at most layers. However, in **AG News** the pattern breaks: B is comparable to the other students, without any clear evidence that would set it apart. Results in both datasets suggest attention-level *cosine similarity* to the teacher is not a reliable signature of distillation.

Results for *Frobenius distance* show that B cannot be told apart from any student without the plot's legend, across every layer on both datasets. All models peak near the theoretical maximum at $i =$ 3–4, then stabilize near half that threshold.

D_{JS} shows a similar growth trend to *Frobenius distance*, but with the sole exception that it does not fully collapse across models. In both datasets, there is a layer i beyond which B is consistently above S_0, S_1, and S_2. This is suggestive evidence

Table II: Jaccard@10, Spearman ρ, and faithfulness@10 vs. teacher for **LIME**, **SHAP**, and **Integrated Gradients** (IG, zero baseline), plus **LIME**-only stability (mean pairwise Jaccard@10 across 5 seeds, $N = 20$), both datasets ($N = 100$, $k = 10$). The **best** and second best results are highlighted.

Model	LIME				SHAP			IG		
	Jac.@10	ρ	Faithf.	Stab.	Jac.@10	ρ	Faithf.	Jac.@10	ρ	Faithf.
IMDb										
P	–	–	0.172	0.548	–	–	0.118	–	–	0.010
B	0.183	0.248	0.083	0.587	0.117	0.239	0.025	0.793	−0.005	0.010
S_0	0.170	0.216	0.202	0.684	0.123	0.238	0.093	0.778	0.028	**0.085**
S_1	0.136	0.189	0.177	0.611	0.096	0.245	0.058	**0.814**	0.010	0.025
S_2	**0.212**	**0.299**	**0.267**	**0.710**	**0.176**	**0.333**	**0.120**	0.787	**0.039**	0.019
AG News										
P	–	–	0.138	**0.773**	–	–	**0.002**	–	–	0.053
B	0.336	0.321	0.306	0.668	0.315	0.269	−0.002	**0.890**	0.021	0.067
S_0	0.347	0.279	0.402	0.638	0.317	**0.274**	0.000	0.883	**0.040**	0.162
S_1	0.341	0.306	0.299	0.614	**0.320**	0.267	0.000	0.876	−0.036	0.138
S_2	**0.358**	**0.342**	**0.431**	0.688	0.311	0.262	−0.000	0.888	0.013	**0.340**

for a genuine effect of distillation over the student's attention maps. However, it would require testing across more datasets and tasks to confirm it.

PR values of P_{avg} remain close to 1 in both datasets for all layers, while students grow substantially away from it. On **IMDb**, B and all students remain indistinguishable without the plot's legend. However, on **AG News** B separates considerably from the other three, though without any clear reason why; since PR does not behave the same way across datasets, it cannot be read as general evidence of a KD effect, though the **AG News** separation is notable enough that a distillation-specific effect there cannot be ruled out either. ER keeps all four models relatively close together, with the one exception being S_2 on **AG News** – not B, which shows little distinctness in these plots. Spectral complexity, then, offers no dataset-consistent echo of the D_{JS} pattern.

F. Discussion

Taken together, these results indicate that, at least in the studied settings, logit-based KD does not directly induce alignment of either explanatory mechanisms or most internal representations properties between teacher and student. Distillation's only provable effect is distributional: it reliably pulls the student's output distribution toward the teacher's (Table I), while accuracy itself is barely, and inconsistently, affected. Explanatory similarity, and most representational metrics, instead appear to emerge mostly from the shared dataset and task rather than from the distillation itself.

If explanatory or representational alignment were induced by the distillation term, both should track distillation strength directly. On **IMDb**, S_1 attains *lower* D_{KL} than S_2 (Table I) yet is markedly less **LIME/SHAP**-aligned with the teacher, and on **AG News**, B matches or beats the distilled students in attention-level *cosine similarity* to the teacher (Fig. 1) despite receiving no distillation signal at all. The one partial exception is D_{JS}, which consistently separates B from the distilled students on both datasets – a hint that distillation may have some attention-level influence, though confirming it would require testing beyond these two datasets and tasks.

G. Limitations

This study is limited to two English datasets for text classification with modest label cardinality; whether the observed task-dependence generalizes to other modalities or to higher-cardinality tasks remains untested. The (α, β) grid is narrow: only two points beyond S_1/S_2 were tested ($\alpha = 1$ for B, $\alpha = 0.9$ for S_0), and the opposite extreme (pure distillation, $\alpha = 0$) was never tested. Underlying all of this is a hardware constraint: every experiment ran on a single consumer GPU (RTX 5070 Ti, 16GB), bounding how many datasets, scales, and architectures could be explored.

IV. Conclusions

This work investigated whether functional similarity induced by knowledge distillation extends to internal representations and explanatory mechanisms, on binary **IMDb** sentiment classification and four-class **AG News** topic classification. In these comparatively simple settings, the answer is negative rather than definitively so. The one property that behaves consistently, and scales with distillation strength, independently of accuracy, is distributional fidelity to the teacher's output; neither explanatory mechanisms nor most representational metrics follow it.

On explanatory mechanisms, the dissociation is clearest on **IMDb**: S_1 reproduces the teacher's output distribution more faithfully than S_2 (lower D_{KL}), yet is markedly *less* **LIME/SHAP**-aligned with the teacher – the opposite of what a distillation-driven account predicts. Every student, including the non-distilled B, assigns importance to tokens with no topical or sentiment content regardless of its distillation weight, so this failure to transfer explanatory mechanisms cannot be attributed to insufficient distillation strength either.

A striking pattern underlies the representational side of this: B, S_0, S_1, and S_2 are largely indistinguishable from one another on most metrics – not because their attention structures

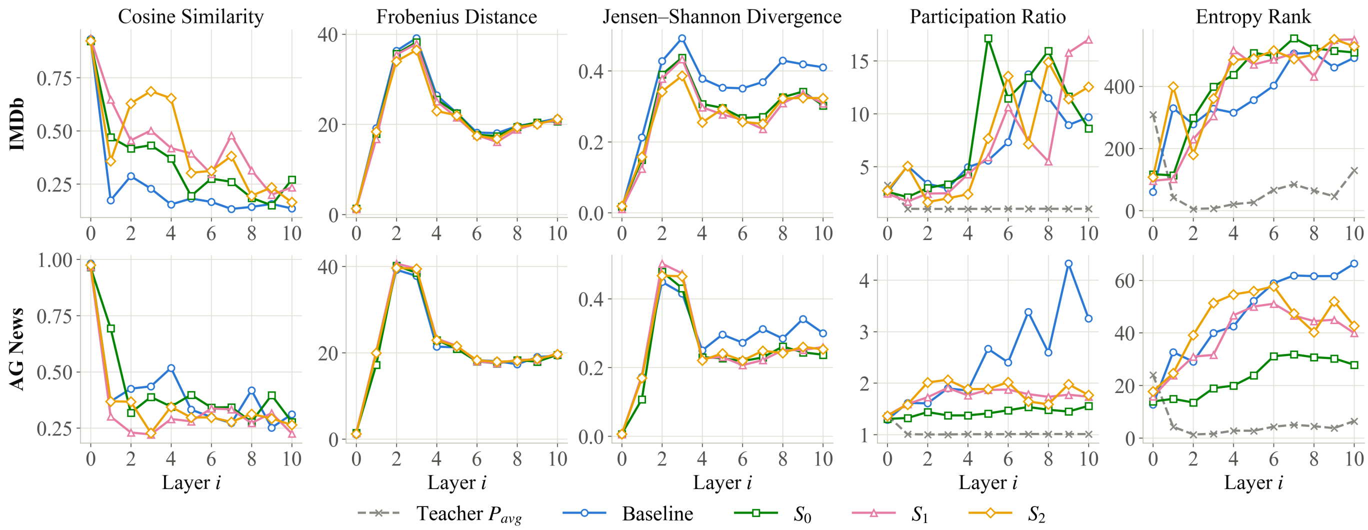

Figure 1: *Cosine similarity*, *Frobenius distance*, row-averaged *Jensen–Shannon divergence*, participation ratio, and entropy-based effective rank between each model's attention maps and the teacher's P_{avg}, both averaged over the test set, for the non-distilled baseline (B) and the distilled students (S_0, S_1, S_2), on both datasets **IMDb** and **AG News**.

are shown to be similar, but because no metric separates them consistently across both datasets, despite spanning α from 0.5 to 1. D_{JS} is the one exception, consistently ranking B as the most divergent from the teacher on both datasets. Since B receives no distillation signal at all, this near-uniformity cannot be attributed to distillation; a natural next step is a student trained with *pure* distillation, which would help disentangle cross-entropy training over the shared dataset from distillation itself as the source of this convergence.

Acknowledgment

The authors extend their sincere gratitude to Mauricio Velasco for his valuable feedback. The writing of this manuscript was partially assisted by generative language models used exclusively for stylistic revision and textual reformulation tasks: Claude Sonnet 4.6 [19] (Anthropic).

References

[1] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[2] Sarthak Jain and Byron C Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556, 2019.

[3] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International journal of computer vision*, 129(6):1789–1819, 2021.

[4] Yang Zhou, Xu Gao, Zichong Chen, and Hui Huang. Attention distillation: A unified approach to visual characteristics transfer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 18270–18280, 2025.

[5] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. " why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1135–1144, 2016.

[6] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.

[7] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR, 2017.

[8] Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 11–20, 2019.

[9] Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*, 2024.

[10] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[11] Timo Schick and Hinrich Schütze. Exploiting cloze-questions for few-shot text classification and natural language inference. In *Proceedings of the 16th conference of the European chapter of the association for computational linguistics: main volume*, pages 255–269, 2021.

[12] Andrew Maas, Raymond E Daly, Peter T Pham, Dan Huang, Andrew Y Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*, pages 142–150, 2011.

[13] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. *Advances in neural information processing systems*, 28, 2015.

[14] Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT press Cambridge, 2016.

[15] Jianhua Lin. Divergence measures based on the shannon entropy. *IEEE Transactions on Information theory*, 37(1):145–151, 1991.

[16] Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *2007 15th European signal processing conference*, pages 606–610. IEEE, 2007.

[17] Stefano Recanatesi, Serena Bradde, Vijay Balasubramanian, Nicholas A Steinmetz, and Eric Shea-Brown. A scale-dependent measure of system dimensionality. *Patterns*, 3(8), 2022.

[18] Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C Wallace. Eraser: A benchmark to evaluate rationalized nlp models. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4443–4458, 2020.

[19] Anthropic. Claude sonnet 4.6 [Large language model], 2025. Used as a generative AI tool for language editing and writing assistance.